Chapter 3 – Agile Methodology

Learning Objectives

Understand Agile principles, values, Scrum framework, Agile lifecycle, roles, ceremonies, artifacts, estimation techniques, and real-world Agile implementation.

What is Agile?

Agile is an iterative and incremental software development approach that delivers value in small, frequent releases while embracing changing requirements through continuous customer collaboration.

Why Agile?

Traditional Development	Agile Development
Sequential	Iterative & Incremental
Late feedback	Continuous feedback
Fixed scope	Adaptive scope
One major release	Frequent releases
Heavy documentation	Working software focus

Agile Manifesto

Values

• Individuals and interactions over processes and tools
• Working software over comprehensive documentation
• Customer collaboration over contract negotiation
• Responding to change over following a plan

Agile Lifecycle

```
Product Vision
      |
Product Backlog
      |
Sprint Planning
      |
+----------------------+
| Sprint (2–4 weeks)   |
| Design               |
| Develop              |
| Test                 |
| Daily Scrum          |
+----------------------+
      |
Sprint Review
      |
Retrospective
      |
Next Sprint
```

Scrum Roles

Product Owner – Maximizes product value and manages backlog.
Scrum Master – Facilitates Scrum and removes impediments.
Developers – Build, test, and deliver working software.

Scrum Events

Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective, and the Sprint itself.

Scrum Artifacts

Product Backlog, Sprint Backlog, Increment, Definition of Done.

User Story Format

As a <role>, **I want** <feature>, **so that** <benefit>.

Agile Estimation

Teams estimate work using Story Points, Planning Poker, T-shirt sizing, and Velocity.

Advantages

• Faster delivery
• Continuous customer feedback
• Better quality
• Lower project risk
• Improved adaptability

Challenges

• Requires active stakeholder involvement
• Scope may evolve frequently
• Team discipline is essential

Real-world Case Study

An online food delivery platform releases features incrementally: Sprint 1 delivers login and registration, Sprint 2 adds restaurant search, Sprint 3 introduces payments, and Sprint 4 enables live order tracking based on continuous customer feedback.

Interview Questions

1. Explain Agile values and principles.
2. Scrum vs Agile.
3. What is a Sprint?
4. Explain Product Backlog vs Sprint Backlog.
5. What is Velocity?
6. What happens during a Sprint Retrospective?